SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I S.Roshan (191126004) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	0	1
Explanation	25%	1	1	0	1
Application	20%	1	1	0	1
Organization	15%	1	1	0	1
Accuracy	10%	1	1	0	1
Clarity	5%	1	1	0	1

Total Marks 20 **/40**

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of

3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Sowal no : 29

SIMATS ENGINEERING

ASSESSMENT TOOL 1

CO - 1

Course Name : Computer Networks
Course Code : CSA0709

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1) File Transfer – Transport Layer and Data Link Layer.

Given:

File Size = 150MB

Network Speed = 50 Mbps

Segment Size = 1500 bytes.

Frame overhead = 40 bytes

Step 1: Convert File Size into bytes.

Assuming 1MB = 10^6 bytes.

$150\ MB = 150 \times 10^6 = 150,000,000$ bytes.

Step 2: Calculate the number of segment

$$\text{Segments} = \frac{\text{File Size}}{\text{Segment Size}}$$

$$= \frac{150,000,000}{1500} = 100,000 \text{ segments.}$$

Step 3: Calculate the total transmission overhead

Overhead = $100,000 \times 40$

$4,000,000$ bytes = 4MB.

Step 4: Calculate total transmission time.

$150,000,000 + 4,000,000 = 154,000,000$ bytes.

Convert bytes to bits.

$154,000,000 \times 8 = 1,232,000,000$ bits.

Transmission time:

$$T = \frac{\text{Total bits}}{\text{Data rate}}$$

$$T = \frac{1,232,000,000}{50,000,000}$$

$T = 24.64$ seconds.

2) Sliding window protocol

Given:

* sliding window Size = 6 Frames
* User data per Frame = 1024 bytes.
* Total Frames = 48
* one Frame in every window is lost and must be retained.

Step 1:

$$\text{Window} = \frac{\text{Total Frames}}{\text{window size}}$$

$$= \frac{48}{6}$$

$$= 8 \text{ windows}$$

Step 2:

One Frame is lost in every window.

Retransmission = 8 x1

8 retransmission

Step 3: ~~Step 3:~~ calculate the total no. of Frames transmitted.

Total transmission = 48 + 8

Step 4: calculate the total user data.

56 Frames.

48 x 1024 = 49,152 bytes.

49,152 bytes.

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3) IP packet Fragmentation:

Given

IP packet Size = 12,000 bytes.

MTU = 1500 bytes.

IP header per Fragment = 20 bytes.

Step 1:

Data per Fragment = MTU - Header

= 1500 - 20

1480 bytes

Step 2:

No. of Fragment = $\left\lceil \frac{12000}{1480} \right\rceil$

= $\lceil 8.1081 \rceil$

9 Fragment.

Step 3:

Each Fragment has a 20-byte IP header

Header Overhead = 9 x 20

180 bytes.

Sliding Window protocol:

given:

Sliding window size = 6 Frames

Data per Frame = 1024 bytes

Total Frames = 48

one Frame in every window is lost

Step 1: Calculate the number of windows

$$\text{Windows} = \frac{48}{6}$$

8 windows.

Step 2: Calculate the Retransmission

one frame is lost in each window

Retransmission = 8 × 1

8 frame.

Step 3:

Total transmission = 48 + 8

56 Frames.

Step 4:

48 × 1024 = 49,152 bytes

~~49500 bytes~~ . 49,152 bytes.